I. MCQ

1. D	6. A	11. D	16. C
2. A	7. B	12. B	17. B
3. D	8. C	13. B	18. B
4. A	9. D	14. C	19. A
5. C	10. B	15. A	20. C

II. Answer the questions:

21. Supervised learning is a type of machine learning algorithm that takes one or more input variables to predict the outcome. It is trained on labeled data. The common supervised learning methods are Nearest ~~Neigh~~ Neighbor, Naive Bayes, Decision Tree, Linear regression...

28. Data-preprocessing is the first and most important step in machine learning. The quality of the outcome is good depends on ~~the~~ data-preprocessing. ~~Data~~ Data-proprocessing is a method which the data scientist use to check for relationship and structure of the data, removing ~~or tim~~ trimming null values, changing data type, Then, sample and reduce data dimension, transform data from low to high level, adding and enhancing the data and finally split data into 2 for training and testing.

25. 2 major applications of data science:

- Machine learning: it is used to train a model or an agent to predict, classify or perform a certain task. Example: ChatGPT from OpenAI
- Data Analysis: it is used in Big Data or usually in financial sector which allow the users to gather many data and depict or represent it in graphical to shows patterns.

29. K-NN is an algorithm which gathers all of the possibles cases or variables then classifies as new case based on the similarity. Therefore, normalization is required because to see the similarity, it must goes through some techinques to calculate and put in a specific range like min-max regression and Z-score.

27. To find an Outlier in the Data:

- plot the data
- find the unusual or abnormal data points
- calculate the median of the point or use boxplot
- if a point way to far from the middle, it is an outlier

30. To discretize the continuous attributes:

+ ~~Thi~~ An ~~ext~~ example of this is the time-series data. You must split the the timeframe into small ~~pr~~ periods and categorize it.

~~Example: Date: 01-01-2023 .... 31-01-2023~~

~~⇒ B Date s~~

If the data contains, date and times, then classify it to only date and categorize the time to a period.

26. Structured vs Semi-Structured data:

- Structured data are ~~oe~~ organized data which usually stores in tabular or JSON format with proper variables or column names and hierarchy.
- Semi-Structured data are ~~also~~ ~~org~~ less organized data in a form of XML. It ~~is~~ has proper tag to classify ~~it s~~ itself as something, but has no exact format to store data like JSON or rows and columns.

33. Gradient deseent is a multiplicative number which show the direction of a line in a function.

24. Two challenges of Machine Learning:

- Data validation: when the model has finished every steps of preprocessing and begin ~~send~~ to use Howerer, the testing performance is low due to mistakes in pre-processing steps.
- Data cleansing: It needs alot of work and concentration to find outliers, change format or data type replace special characters, removing and trimming data.

2

31. K in K-means is used to separate the partition of the class.

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